



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus

ACADEMIC GRADUATE STUDIES AND RESEARCH DIVISION

FIRST SEMESTER 2026-27
COURSE HANDOUT (Part-II)

Date: Aug 03, 2026

In addition to the part I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course –

Course No. : BITS F327
Course Title : Artificial Intelligence for Robotics
Instructor-in-Charge : Dr. Yogesh Singh (yogesh.singh@pilani.bits-pilani.ac.in)
Google Classroom Code : dubb6k34

1. Course Description:

This course provides a comprehensive introduction to artificial intelligence techniques for robotic systems with emphasis on adaptive learning, machine learning, and cognitive decision making. Topics include: Evolution of intelligent robotics; adaptive learning algorithms; biologically inspired learning using Dynamic Hebbian Learning and adaptive oscillators; statistical learning and adaptive estimation; data-driven modelling of serial-chain manipulator kinematics using neural networks; reinforcement learning for robot trajectory planning and decision making; and high-level robot control through large language models/vision language models (using camera sensors). Each module contains hands-on sessions and project-based learning to emphasize implementation of intelligent algorithms for adaptation, learning, and autonomous algorithm for robotics.

2. Scope and Objective of the Course:

This course introduces the fundamental concepts of Artificial Intelligence for Robotics through the lens of adaptive, data-driven, and cognitive robotic systems. Unlike conventional robotics courses that primarily focus on analytical modelling and control, this course emphasizes how robots can learn, adapt, and make intelligent decisions using modern AI techniques while remaining grounded in classical robotics principles.

Students will explore multiple paradigms of robot intelligence, ranging from adaptive learning algorithms and biologically inspired learning mechanisms to neural networks, reinforcement learning, and high-level cognitive reasoning using vision and large language models. Through hands-on experiments, students will investigate how intelligent algorithms can be applied to serial-chain robotic manipulators for learning kinematics, trajectory generation, sensor adaptation, and autonomous task execution.

Course Outcome:

Upon successful completion of this course, students will be able to:

CO1	Explain and compare adaptive, statistical, machine learning, and AI-based approaches for developing intelligent robotic systems.
CO2	Develop and implement adaptive learning algorithms, including Dynamic Hebbian Learning and statistical learning methods, for robotic sensing, adaptation, and motion generation.
CO3	Design and evaluate neural network and reinforcement learning approaches for solving robot kinematics and trajectory planning problems in serial-chain manipulators.
CO4	Develop AI-enabled robotic applications that integrate vision to generate high-level task commands for autonomous robotic manipulation.



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3. Textbook(s):

- (T1) Frank, C. (2017). Modern robotics-mechanics, planning, and control. Cambridge University Press.
(T2) Righetti, L., Buchli, J., & Ijspeert, A. J. (2006). Dynamic hebbian learning in adaptive frequency oscillators. *Physica D: Nonlinear Phenomena*, 216(2), 269-281.
(T3) Vaseghi, S. V. (2008). Advanced digital signal processing and noise reduction. Hoboken.
(T4) Aggarwal, C. C. (2018). Neural networks and deep learning (Vol. 10, No. 978, p. 3). Cham: springer.

4. Reference Book(s):

- (R1) Lung-Wen Tsai, "Robot Analysis – The mechanics of serial and parallel Manipulators", John Wiley & Sons, Inc
(R2) Sciavicco, L., & Siciliano, B. (2012). *Modelling and control of robot manipulators*. Springer Science & Business Media.
(R3) Richard S. Sutton and Andrew G. Barto, Reinforcement Learning: An Introduction.

5. Lecture Plan:

Module #	Lecture Session	Reference	Learning Outcome
M1	Foundations for Terminology in AI and Robotics (M1) Lectures 1 to 3 – Introduction to AI in Robotics, Evolution of robot intelligence, AI/ML vs Model-based vs adaptive vs learning algorithms, Basic robot terminology and nomenclature.	T1	Explain, differentiate, and analyze the fundamental concepts of artificial intelligence, adaptive learning, and intelligent robotic systems
M2	Adaptive Learning in Dynamical Systems (M2) Lectures 4 to 12 – Biological inspiration, central pattern generators, Hebbian Learning (dynamic), Learning rhythmic motion, effect of perturbations, applications in walking robots, exoskeletons, and prosthetic systems. Hands-On Experiment/Practical Project: Apply dynamic Hebbian learning algorithm to real-life periodic signals generated from IMUs and/or Encoders.	T2, Class Notes, Research articles	Analyze and implement biologically inspired adaptive learning algorithms for rhythmic motion generation and adaptive robotic systems.
M3	Adaptive Statistical Learning for Robotics (M3) Lectures 13 to 20 – Least square Error Wiener-Kolmogorov Filters, Derivation of Wiener Filter Equation, Autocorrelation of Input and Cross-Correlation of I/P and Desired signals, QR Decomposition, Interpretation of Wiener Filters as projections in vector space, Analysis of LMS Error Signal, Applications Hands-On Experiment/Practical Project: Apply statistical based learning on real-time IMU data and/or Case Study on Kalman Filtering with IMU.	T2, Class Notes, Research articles	Formulate and implement, statistical learning algorithms, for adaptive estimation, sensor data processing, and robotic signal analysis.
M4	Data-Driven Machine Learning approach in Kinematics and Trajectory Planning of Serial-Chain Manipulators (M4) Lectures 21 to 35 – Forward/Inverse Kinematics and Trajectory Planning in SCM (Conventional way), Data-driven modeling using Neural Networks for inverse kinematics in SCM. Reinforcement Learning approach in Trajectory Planning	T1, T4, R3, Class Notes, Research articles	Design, develop, and evaluate neural network and reinforcement learning models for robot kinematics, trajectory planning, and intelligent motion generation in robotic SCM.





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	Hands-On Experiment/Practical Project: Inverse kinematics and trajectory planning using Neural Network and Reinforcement Learning on a 3R spatial robotic manipulator.		
M5	High-Level Robot Task Planning and Execution (M5) Lectures 36 to 42 – Vision-to-action primitives, LLMs/VLMs based planning for kinematics and trajectory planning in SCM. Hands-On Experiment/Practical Project: Use Camera Sensor and LLMs/VLMs on 3R spatial SCM to execute a task in the robot's workspace.	Class Notes, Class Experiments	Develop and integrate large language models (LLMs) and vision-language models (VLMs) to generate and execute high-level task commands with simple robotic SCM.

6. Evaluation Scheme:

Component	Wt (%)	Duration	Date & Time	Comments
Hand-On Projects / Case Studies	45	TBA	Post each module	- Perform in a team of maximum 4 students. - Best 3 of 4
Mid Semester Exam	20	90 min(s)	As announced in the timetable	Closed Book
Comprehensive Exam	35	180 min(s)	As announced in the timetable	Open Book and/or partly Software Based (MATLAB, Python, Robot Software, etc.)

Note - Minimum Passing Marks Criteria:

The passing marks for this course will be determined based on one of the criteria - 30% of the median of the post-comprehensive examination score or 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as a passing mark in this course. Failure to meet this criterion will result in an NC grade.

7. Office Consultation Hour:

Announced or by prior appointment via email. Reach out to the IC at yogesh.singh@pilani.bits-pilani.ac.in
Office Location: NAB 6020-D.

8. Make-up Policy

Make-up will be granted only in genuine cases with prior permission as per the institute's rules.

9. Attendance Policy

Attendance is not mandatory. However, students are strongly encouraged to attend lectures, actively engage in class discussions, and benefit from the learning experience.

Instructor-In-Charge
BITS F327: AI for Robotics



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